

CADET _____ SECTION _____ TIME OF DEPARTURE _____

DEPARTMENT OF CHEMICAL & BIOLOGICAL SCIENCE & ENGINEERING

CH365 2026-2027

TEXT: Smith, Van Ness, & Abbott

WRITTEN PARTIAL REVIEW I

SCOPE: Lessons 1-8

10 September 2026, A-Hour

References Permitted: Open note, book, and computer. You may not share files or communicate with other cadets in any way during the exam.

INSTRUCTIONS

1. You will have 55 minutes to complete the exam.
2. Do not mark the exam or open it until “begin work” is given.
3. There are 3 problems on 5 pages (not including the cover page). Print your name on the top of each sheet. Answer all questions.
4. Solve the problems in Mathematica or in the space provided. Show work to receive for partial credit.

(TOTAL WEIGHT: 200 POINTS)

DO NOT WRITE IN THIS SPACE

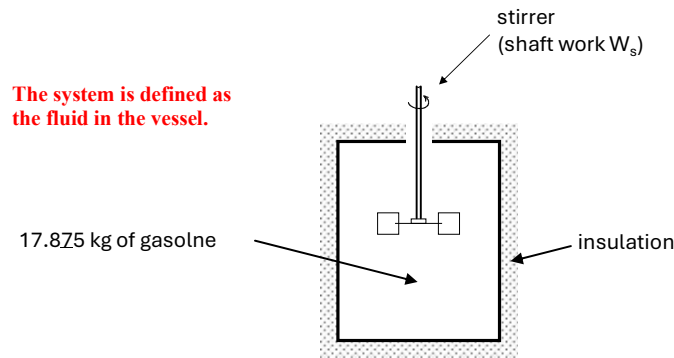
PROBLEM	VALUE	CUT
A	60	
B	70	
C	70	
TOTAL CUT		
TOTAL GRADE	200	

Problem: Weight:
A 60

Joule Experiment: An unsealed but well-insulated container filled with 17.875 kg of gasoline at 15.330 °C is fitted with a stirrer made to turn by gravity acting on a weight of mass 50.00 kg connected to the stirrer by a cable and a system of pulleys. Initially, the weight and the fluid are completely at rest. When the weight drops, energy is transferred through the pulley system into the gasoline through the stirrer. After the weight stops falling, the gasoline is allowed to come to rest so that any swirling motion ceases and the bulk velocity of the fluid is zero. At this point, the weight and the fluid are again completely at rest, but the temperature is observed to have increased, leading to Joule's hypothesis that the potential energy change of the weight is transferred to the gasoline.

The first law for a steady-state, steady flow process is given by equation 2.31 in the textbook. Note carefully that this equation also applies to non-flowing systems because the flow rate was subsumed into the different terms. The local acceleration of gravity is 9.817 (m)(s⁻²), and the specific heat of gasoline is 2.22 (kJ)(kg⁻¹)(°C⁻¹).

(a) Define the thermodynamic system in a way that allows direct application of equation 2.31. Provide a sketch of the system.



(b) Write and simplify equation 2.31 for the Joule experiment as described above, in a manner consistent with your answer to part (a). Explain any simplifying assumptions.

Solution:

$$\Delta H + \frac{\Delta(u^2)}{2} + g\Delta z = Q + W_{\text{shaft}} \quad (\text{Equation 2.31})$$

The system (fluid) is at rest before and after the weight drop. $\Delta(u^2) = 0$

The fluid does not change elevation. $g\Delta z = 0$

There is no heat transfer to or from the surroundings. $Q = 0$

$$\underline{\underline{\Delta H = W_{\text{shaft}}}} \quad \text{ANS}$$

(c) Use the definition of enthalpy to explain how ΔH in equation 2.31 leads to ΔU and explain any additional simplifying assumptions needed.

Solution:

- Since the experiment is open to the atmosphere, pressure is assumed constant.
- Molar volume of the oil is assumed constant for small temperature changes.

$$\Delta H = \Delta U + \Delta(PV) = \Delta U$$

$$\therefore \underline{\underline{\Delta H = \Delta U}}_{\text{ANS}}$$

(d) Determine the internal energy change of the oil if the weight drops a total of 60.00 m. Report your answer in kJ.

Solution:

- Assume all potential energy of the falling weight is transmitted into the fluid through the stirring mechanism.
- The change in potential energy of the weight is equal but opposite in sign to the change in internal energy of the fluid.

$$\Delta U = W_{\text{shaft}} = -mg\Delta z = -50.00 \text{ kg} \cdot 9.81 \frac{\text{m}}{\text{s}^2} \cdot (0 \text{ m} - 60.00 \text{ m}) \cdot \frac{1 \text{ J}}{1 \text{ kg} \cdot \text{m}^2 / \text{s}^2} \cdot \frac{1 \text{ kJ}}{1000 \text{ J}} = 29.45 \text{ kJ}$$

ANS

(e) What is the final temperature of the oil in °C?

Solution:

- ΔH is given by eq. 2.21 with constant C_p , which leads to $\Delta H = mC_p\Delta T (= \Delta U)$:

$$\Delta H = m \cdot C_p \cdot \Delta T = \Delta U$$

$$29.45 \text{ kJ} = 17.875 \text{ kg} \cdot \frac{2.22 \text{ kJ}}{\text{kg} \cdot ^\circ\text{C}} \cdot (T \text{ } ^\circ\text{C} - 15.330 \text{ } ^\circ\text{C})$$

$$\underline{\underline{T = 16.072 \text{ } ^\circ\text{C}}}_{\text{ANS}}$$

Problem: Weight:
B 70

1. A mercury manometer is attached at one end to a pressure vessel containing a gas and is open to the atmosphere at the other end. The reading on the manometer is 27.53 inches of mercury (in Hg) and the atmospheric pressure is 29.96 in Hg. Use the additional information given below to find the absolute pressure in the vessel in psia.

Additional information: The temperature of the vessel and the surroundings are both 70 °F, the density of mercury at 70 °F is 13.543 g/cm³, the local acceleration of gravity is 32.254 ft/s², and 1 torr = 1 mm Hg. Additional conversion factors can be found in Appendix A.

Solution:

$$P_{\text{absolute}} = P_{\text{gauge}} + P_{\text{atm}} = 27.53 + 29.96 = 57.49 \text{ in Hg}$$

$$\frac{13.543 \text{ g}}{\text{cm}^3} \cdot \frac{2.20462 \text{ lb}_m}{1000 \text{ g}} \cdot \frac{10^6 \text{ cm}^3}{35.3147 \text{ ft}^3} = 845.4601 \frac{\text{lb}_m}{\text{ft}^3}$$

$$\frac{\left(\frac{845.4601 \text{ lb}_m}{\text{ft}^3} \right) \cdot \left(\frac{32.254 \text{ ft}}{\text{s}^2} \right) \cdot (57.49 \text{ in}) \cdot \left(\frac{1 \text{ ft}}{12 \text{ in}} \right)}{\frac{32.1740 \text{ ft} \cdot \text{lb}_m}{\text{s}^2} \cdot \frac{1 \text{ lb}_f}{1 \text{ lb}_m}} \cdot \left(\frac{1 \text{ ft}}{12 \text{ in}} \right)^2 = \frac{28.198 \text{ lb}_f}{\text{in}^2} = 28.20 \text{ psia}$$

ANS

2. A mass of 5.0 kg is placed on a dead-weight gauge with a piston diameter of 10.00 mm. What is the pressure produced by gauge in bar? The local acceleration of gravity is 9.811 m/s².

Solution:

Pressure is force per area, so calculate the force, calculate the area from the given diameter, then divide:

$$\frac{5.0 \text{ kg} \cdot \frac{9.811 \text{ m}}{\text{s}^2} \cdot \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m}}}{\frac{\pi}{4} \left(10.00 \text{ mm} \cdot \frac{1 \text{ m}}{1000 \text{ mm}} \right)^2} \cdot \frac{1 \text{ bar}}{10^5 \text{ N} / \text{m}^2} = 6.246 \text{ bar}$$

ANS

Problem: Weight:
C 70

One kilogram of hydrogen sulfide is expanded reversibly and isothermally from an initial state of 2.50 L/mol (State 1) until its molar volume increases to 35.0 L/mol and its pressure decreases to 2.00 bar (State 2).

The table summarizes the conditions at states 1 and 2, where the question marks (??) designate unknowns. Complete the table below and then calculate W , Q , ΔH , and ΔU for the process, in units of kJ mol^{-1} .

Assume that $PV/T = 0.08314 \text{ bar L mol}^{-1} \text{ K}^{-1}$ and that $C_p = 36.26 \text{ J mol}^{-1} \text{ K}^{-1}$ for hydrogen sulfide.

	State 1	State 2
P, bar	<u>28.0</u> ??	<u>2.00</u>
T, K	<u>842.0</u> ??	<u>842.0</u> ??
V, L/mol	<u>2.50</u>	<u>35.0</u>

Solution:

Solve for T_2 :

$$\frac{P_2 V_2}{T_2} = 0.08314 \Rightarrow T_2 = \frac{P_2 V_2}{0.08314} = \frac{2.00 \text{ bar} \cdot 35.0 \frac{\text{L}}{\text{mol}}}{0.08314 \frac{\text{bar} \cdot \text{L}}{\text{mol} \cdot \text{K}}} = 84 \underline{1.95} \text{ K}$$

The system is at constant temperature, so $T_1 = T_2 = \underline{\underline{841.95 \text{ K}}}$ ANS

Now solve for P_1 :

$$\frac{P_1 V_1}{T_1} = 0.08314 \Rightarrow P_1 = \frac{T_1 \cdot 0.08314}{V_1} = \frac{84 \underline{1.95} \text{ K} \cdot 0.08314 \frac{\text{bar} \cdot \text{L}}{\text{mol} \cdot \text{K}}}{2.50 \frac{\text{L}}{\text{mol}}} = \underline{\underline{28.0 \text{ bar}}}$$
 ANS

Enthalpy change: $\Delta H = 0 \frac{\text{kJ}}{\text{mol}}$ (isothermal)

ANS

Internal energy change: $\Delta U = 0 \frac{\text{kJ}}{\text{mol}}$ (isothermal)

ANS

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Reversible work done by the system:

$$W = -\int_{V_1}^{V_2} P dV = -\int_{V_1}^{V_2} \frac{RT}{V} dV = -\int_{2.50}^{35.0} \frac{0.08314 \cdot 841.95}{V} dV = -184.73 \cdot \frac{\text{bar} \cdot \text{L}}{\text{mol}}$$

Convert this to kJ/mol:

Note: work done on the system is negative for an expansion.

$$\frac{-184.733 \text{ bar} \cdot \text{L}}{\text{mol}} \cdot \frac{10^5 \text{ Pa}}{\text{bar}} \cdot \frac{1 \text{ N}}{\text{m}^2} \cdot \frac{1 \text{ m}^3}{1000 \text{ L}} \cdot \frac{1 \text{ J}}{1 \text{ Nm}} \cdot \frac{1 \text{ kJ}}{1000 \text{ J}} = -18.473 \frac{\text{kJ}}{\text{mol}}$$

ANS

Heat added to system:

$$\Delta U = Q + W \Rightarrow Q = \Delta U - W = 0 \frac{\text{kJ}}{\text{mol}} - (-18.473) \frac{\text{kJ}}{\text{mol}} = 18.473 \frac{\text{kJ}}{\text{mol}}$$

ANS

Note: heat is positive.

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4. Solve the problems in Mathematica or in the space provided. Show work to receive for partial credit.

(TOTAL WEIGHT: 200 POINTS)

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B	70	
C	70	
TOTAL CUT		
TOTAL GRADE	200	

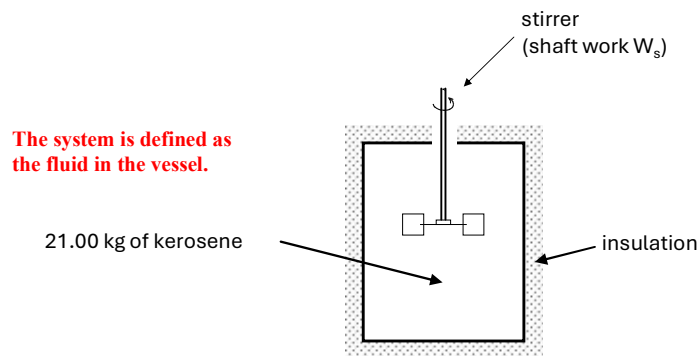
Problem: Weight:
A 60

Joule Experiment: An unsealed but well-insulated container filled with 21.00 kg of kerosene at 14.130 °C is fitted with a stirrer made to turn by gravity acting on a weight of mass 75.00 kg connected to the stirrer by a cable and a system of pulleys. Initially, the weight and the fluid are completely at rest. When the weight drops, energy is transferred through the pulley system into the kerosene through the stirrer. After the weight stops falling, the kerosene is allowed to come to rest so that any swirling motion ceases and the bulk velocity of the fluid is zero. At this point, the weight and the fluid are again completely at rest, but the temperature is observed to have increased, leading to Joule's hypothesis that the potential energy change of the weight is transferred to the kerosene.

The first law for a steady-state, steady flow process is given by equation 2.31 in the textbook. Note carefully that this equation also applies to non-flowing systems because the flow rate was subsumed into the different terms. The local acceleration of gravity is 9.793 (m)(s⁻²), and the specific heat of kerosene is 2.01 (kJ)(kg⁻¹)(°C⁻¹).

(a) Define the thermodynamic system in a way that allows direct application of equation 2.31. Provide a sketch of the system.

Solution:



(b) Write and simplify equation 2.31 for the Joule experiment as described above, in a manner consistent with your answer to part (a). Explain any simplifying assumptions.

Solution:

$$\Delta H + \frac{\Delta(u^2)}{2} + g\Delta z = Q + W_{\text{shaft}} \quad (\text{Equation 2.31})$$

The system (fluid) is at rest before and after the weight drop. $\Delta(u^2) = 0$

The fluid does not change elevation. $g\Delta z = 0$

There is no heat transfer to or from the surroundings. $Q = 0$

$$\underline{\underline{\Delta H = W_{\text{shaft}}}} \quad \text{ANS}$$

(c) Use the definition of enthalpy to explain how ΔH in equation 2.31 leads to ΔU and explain any additional simplifying assumptions needed.

Solution:

- Since the experiment is open to the atmosphere, pressure is assumed constant.
- Molar volume of the oil is also essentially constant for small temperature changes.

$$\Delta H = \Delta U + \Delta(PV) = \Delta U$$

$$\therefore \underline{\underline{\Delta H = \Delta U}}_{\text{ANS}}$$

(d) Determine the internal energy change of the kerosene if the weight drops a total of 40.00 m. Report your answer in kJ.

Solution:

- Assume all potential energy of the falling weight is transmitted into the fluid through the stirring mechanism.
- The change in potential energy of the weight is equal but opposite in sign to the change in internal energy of the fluid.

$$\Delta U = W_{\text{shaft}} = -mg\Delta z = -75.00 \text{ kg} \cdot 9.793 \frac{\text{m}}{\text{s}^2} \cdot (0 \text{ m} - 40.00 \text{ m}) \cdot \frac{1 \text{ J}}{1 \text{ kg} \cdot \text{m}^2 / \text{s}^2} \cdot \frac{1 \text{ kJ}}{1000 \text{ J}} = \underline{\underline{29.379 \text{ kJ}}}_{\text{ANS}}$$

(e) What is the final temperature of the kerosene in °C?

Solution:

- ΔH is given by eq. 2.21 with constant C_p , which leads to $\Delta H = mC_p\Delta T (= \Delta U)$:

$$\Delta H = m \cdot C_p \cdot \Delta T = \Delta U$$

$$29.379 \text{ kJ} = 21.00 \text{ kg} \cdot \frac{2.01 \text{ kJ}}{\text{kg} \cdot ^\circ\text{C}} \cdot (T \text{ } ^\circ\text{C} - 14.130 \text{ } ^\circ\text{C})$$

$$\underline{\underline{T = 14.826 \text{ } ^\circ\text{C}}}_{\text{ANS}}$$

Problem: Weight:
B 70

1. A mercury manometer is attached at one end to a pressure vessel containing a gas and is open to the atmosphere at the other end. The reading on the manometer is 52.23 inches of mercury (in Hg) and the atmospheric pressure is 30.23 in Hg. Use the additional information given below to find the absolute pressure in the vessel in psia

Additional information: The temperature of the vessel and the surroundings are both 70 °F, the density of mercury at 70 °F is 13.543 g/cm³, the local acceleration of gravity is 32.093 ft/s², and 1 torr = 1 mm Hg. Additional conversion factors can be found in Appendix A.

Solution:

$$P_{\text{absolute}} = P_{\text{gauge}} + P_{\text{atm}} = 52.23 + 30.23 = 82.46 \text{ in Hg}$$

$$\frac{13.543 \text{ g}}{\text{cm}^3} \cdot \frac{2.20462 \text{ lb}_m}{1000 \text{ g}} \cdot \frac{10^6 \text{ cm}^3}{35.3147 \text{ ft}^3} = 845.4601 \frac{\text{lb}_m}{\text{ft}^3}$$

$$\frac{\left(\frac{845.4601 \text{ lb}_m}{\text{ft}^3} \right) \cdot \left(\frac{32.093 \text{ ft}}{\text{s}^2} \right) \cdot (82.46 \text{ in}) \cdot \left(\frac{1 \text{ ft}}{12 \text{ in}} \right)}{\frac{32.1740 \text{ ft} \cdot \text{lb}_m}{\text{s}^2} \cdot \frac{1 \text{ lb}_f}{1 \text{ lb}_m}} \cdot \left(\frac{1 \text{ ft}}{12 \text{ in}} \right)^2 = \frac{40.243 \text{ lb}_f}{\text{in}^2} = 40.24 \text{ psi} \quad \text{ANS}$$

Instructor typo – lined out

2. A mass of 20.0 kg is placed on a dead-weight gauge with a piston diameter of 25.00 mm. ~~produces a pressure of 292.65 bar.~~ What is the pressure produced by gauge in bar? The local acceleration of gravity is 9.799 m/s².

Solution:

Pressure is force per area, so calculate the force, calculate the area from the given diameter, then divide:

$$\frac{20.0 \text{ kg} \cdot \frac{9.799 \text{ m}}{\text{s}^2} \cdot \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m}}}{\frac{\pi}{4} \left(25.00 \text{ mm} \cdot \frac{1 \text{ m}}{1000 \text{ mm}} \right)^2} \cdot \frac{1 \text{ bar}}{10^5 \text{ N/m}^2} = 3.992 \text{ bar} \quad \text{ANS}$$

Problem: Weight:
C 70

One kilogram of carbon monoxide is compressed reversibly and isothermally from an initial state of 17.9 L/mol and 2.00 bar (State 1) until its molar volume is reduced to 3.90 L/mol (State 2).

The table below summarizes the conditions at states 1 and 2, where the question marks (??) designate unknowns. Complete the table and then calculate W, Q, ΔH, and ΔU for the process, in units of kJ mol⁻¹.

Assume that PV/T = 0.08314 bar L mol⁻¹ K⁻¹ and that C_p = 29.16 J mol⁻¹ K⁻¹ for carbon monoxide.

	State 1	State 2
P, bar	2.00	9.179 ??
T, K	430.6 ??	430.6 ??
V, L/mol	17.9	3.90

Solution:

Solve for T₁:

$$\frac{P_1 V_1}{T_1} = 0.08314 \Rightarrow T_1 = \frac{P_1 V_1}{0.08314} = \frac{2.00 \text{ bar} \cdot 17.9 \frac{\text{L}}{\text{mol}}}{0.08314 \frac{\text{bar} \cdot \text{L}}{\text{mol} \cdot \text{K}}} = 430.6 \text{ K}$$

The system is at constant temperature, so T₂ = T₁ = 430.6 K.
ANS

Now solve for P₂:

$$\frac{P_2 V_2}{T_2} = 0.08314 \Rightarrow P_2 = \frac{T_2 \cdot 0.08314}{V_2} = \frac{430.6 \text{ K} \cdot 0.08314 \frac{\text{bar} \cdot \text{L}}{\text{mol} \cdot \text{K}}}{3.90 \frac{\text{L}}{\text{mol}}} = 9.179 \text{ bar}$$

ANS

Enthalpy change: $\Delta H = 0 \frac{\text{kJ}}{\text{mol}}$ (isothermal)
ANS

Internal energy change: $\Delta U = 0 \frac{\text{kJ}}{\text{mol}}$ (isothermal)
ANS

This page intentionally left blank for cadet calculations.

Reversible work done on the system:

$$W = -\int_{V_1}^{V_2} P dV = -\int_{V_1}^{V_2} \frac{RT}{V} dV = -\int_{17.9}^{3.90} \frac{0.08314 \cdot 430.599}{V} dV = 54.553 \cdot \frac{\text{bar} \cdot \text{L}}{\text{mol}}$$

Note: work done on the system is positive for a compression.

Convert this to kJ/mol:

$$\frac{54.553 \text{ bar} \cdot \text{L}}{\text{mol}} \cdot \frac{10^5 \text{ Pa}}{\text{bar}} \cdot \frac{1 \text{ N}}{\text{m}^2} \cdot \frac{1 \text{ m}^3}{1000 \text{ L}} \cdot \frac{1 \text{ J}}{1 \text{ Nm}} \cdot \frac{1 \text{ kJ}}{1000 \text{ J}} = 5.4553 \frac{\text{kJ}}{\text{mol}}$$

ANS

Heat removed from system:

$$\Delta U = Q + W \Rightarrow Q = \Delta U - W = 0 \frac{\text{kJ}}{\text{mol}} - 5.455 \frac{\text{kJ}}{\text{mol}} = -5.455 \frac{\text{kJ}}{\text{mol}}$$

ANS

Note: heat is negative.